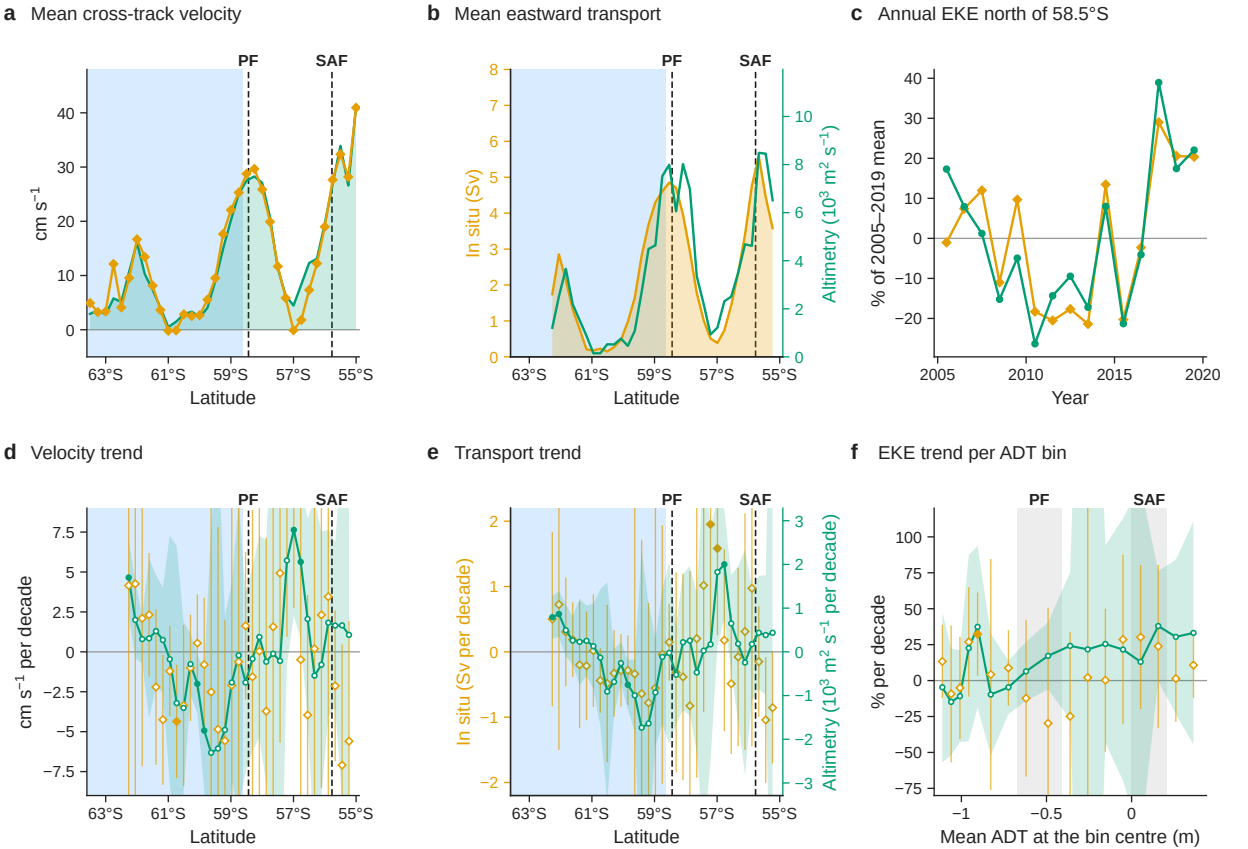
a Mean cross-track velocity
PF
SAF
cm s−1
40
30
20
10
0
63°S
61°S
59°S
57°S
55°S
Latitude
b Mean eastward transport
PF
SAF
In situ (Sv)
Altimetry (10³ m² s⁻¹)
8
7
6
5
4
3
2
1
0
10
8
6
4
2
0
63°S
61°S
59°S
57°S
55°S
Latitude
c Annual EKE north of 58.5°S
% of 2005–2019 mean
40
30
20
10
0
−10
−20
2005
2010
2015
2020
Year
d Velocity trend
PF
SAF
cm s−1 per decade
7.5
5
2.5
0
−2.5
−5
−7.5
63°S
61°S
59°S
57°S
55°S
Latitude
e Transport trend
PF
SAF
In situ (Sv per decade)
Altimetry (10³ m² s⁻¹ per decade)
2
1
0
−1
−2
3
2
1
0
−1
−2
−3
63°S
61°S
59°S
57°S
55°S
Latitude
f EKE trend per ADT bin
PF
SAF
% per decade
100
75
50
25
0
−25
−50
−75
−1
−0.5
0
Mean ADT at the bin centre (m)
in situ (Gutierrez-Villanueva et al., 2023)
altimetry, Product 1
open: p ≥ 0.05
seasonal sea ice